

Haider Kamal

AI & COMPUTER VISION ENGINEER

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PROFILE

Engineer building **production computer-vision and real-time AI systems** that run on-premise, at frame rate, in front of real users: multi-camera surveillance, industrial inspection, live stadium player tracking, digital twins and conversational avatars. Nine years shipping Unity games and authoritative multiplayer before that, which is why the AI work holds a frame budget.

CORE SKILLS

Computer Vision	Object detection (YOLO), multi-object tracking (BoTSORT), appearance re-identification, pose estimation, action recognition, OCR, homography calibration, super-resolution
AI & ML	PyTorch, TensorRT FP16, ONNX Runtime, vision-language models, local LLM serving, fine-tuning, speech-to-text and text-to-speech
Systems	Real-time pipelines, GPU batching and scheduling, hardware decode, latency budgeting, benchmark design, Docker Compose, Redis, WebSocket gateways, FastAPI, on-premise deployment
3D, XR & Games	Unity3D, Gaussian splatting, photogrammetry, image-to-3D, Meta Quest / OpenXR, mobile AR, authoritative multiplayer servers, Blender
Languages	Python, C#, TypeScript / Node.js, SQL

EXPERIENCE

AI & Computer Vision Engineer - Ideofuzion May 2023 - Present
Islamabad, Pakistan. Joined as Sr. Game Developer and moved into applied AI.

- **Sentry Vision, on-premise AI surveillance.** Designed and built a multi-camera theft-detection system running **8-10 RTSP cameras in real time on a \$200-class used GPU**, with no cloud inference and no video leaving the site.
- Architected a **two-stage alert pipeline**: sub-second pose and behaviour heuristics raise the candidate, a local vision-language model confirms it seconds later. Chosen after benchmarking showed a **0.87 inference/s** VLM ceiling made per-frame model calls impossible at any camera count.
- Cut VLM verification cost **~97%** (785 triggers down to 22-69 verifications, measured over a 10-minute 9-camera soak) by grouping detections into episodes with a sliding rejoin window and scene-cut detection.
- Built a GPU benchmark harness ramping synthetic camera fleets against decode, latency and queue-depth gates. It **disproved the team's CPU-bound assumption**: decode held 60/60 fps at 30 cameras while GPU pose saturated at ~26, which rewrote the hardware costing model and caught three deployment bugs before install.
- Eliminated hallucinated weapon detections by re-asking the model a binary question on a zoomed crop and cross-checking against pose evidence: open-ended prompts invented knives from phones, binary framing did not. Also root-caused a missed sub-3s alert target (median **6.75s**) to a serial verify queue and whole-file JSON persistence, not inference.
- **Immersive AR, live stadium player tracking.** Built and deployed the server-side CV pipeline: two 4K RTSP feeds through YOLO detection on TensorRT FP16, BoTSORT tracking with appearance re-ID, homography pitch calibration, team classification and PaddleOCR jersey identification, on a **single NVIDIA L2 at roughly half a GPU per camera**.
- Held per-frame CV latency to **~23 ms** and end-to-end CV-to-AR to **88-153 ms** against a 500 ms p95 target, by running jersey OCR asynchronously in two GPU worker processes so identification never gates frame emission.
- Raised identification reliability with confidence voting, pose-based torso cropping and x2 super-resolution ahead of OCR, and cut identity-swap artefacts with a track-ID stabiliser using speed-scaled re-association, team-colour confirmation and a 12 m/s velocity cap to reject teleports.
- Wrote the capability assessment that **stopped the organisation over-claiming**: replaced an unmeasured "98% accuracy" headline with the measured pair (96-98% merged, 80-92% single-camera), killed an inconsistent sub-200 ms glass-to-glass figure, and flagged a "60,000 concurrent users" claim as infrastructure never load-tested.

OTHER PRODUCTS BUILT AT IDEOFUZION

CementIQ

Automated cement-dispatch audit: bag count, brand, tax track-and-trace code, plate and pilferage checked per truck and reconciled into the plant ERP.

Vision 3D

Photoreal 3D digital twins from a phone video, a drone pass or a single photograph, delivered to web, VR and AR.

Accelerator

AI-native placement-training platform with a coaching AI mentor and VR interview practice on Meta Quest.

Digital Twin Ops

Live 3D twin of a pipeline network with sensor ingest, autonomous drone inspection and predictive integrity scoring.

AvatarAssist

Conversational avatar with a ~2s speech-to-speech round trip, lip-synced on screen, running on the client's own hardware.

EXPERIENCE (CONTINUED)

Game Developer - Ideofuzion

May 2023 - 2024

- Built the **official UIPM Obstacle Laser Run** for Android and iOS, for the world governing body of modern pentathlon: Unity state machines for responsive player control, a custom socket server for real-time multiplayer, friend challenges and leaderboards.
- Engineered **Games-fi**, a play-to-earn platform on Tron with multi-currency in-game earnings and withdrawals, NFT minting and trading, and a secure raffles hub.
- Built **Stunt-Z**, off-road racing across 150+ tracks with advanced vehicle physics, delivered through Google Play Asset Delivery with dynamic biomes synchronised server-side.

Game Developer - Alam Digital HQ

Sep 2021 - Mar 2023

- Led design and development of **Project Zambeel**, a PC open-world third-person story-driven title: the dynamic mission and branching narrative system, high-fidelity asset pipelines, and the performance work to reach AAA-quality visuals.
- Shipped **Valet Parking**, an HD Android parking simulator with realistic physics and a level-based grading system, with asset streaming and memory use optimised for stable frame rates across a wide device range.

Game Developer - Coding Cannon

Jan 2017 - Sep 2021

- Engineered **Operation Basalat**, a team-based Android multiplayer shooter: real-time client-server networking, objective-driven gameplay loops, AI pathfinding for dynamic combat, and the UI feedback and balancing that keeps roles legible in cooperative play.
- Shipped a long tail of independent Android titles, including Rush Hour Run, Pakdam Pakdai, Space Shooter, Hold'em Poker and Paper Boy.

Earlier

2015 - 2020

- **Ecommerce Marketing Manager**, Source House General Trading (2019-2020): regional GCC marketing, and built a responsive e-commerce site with payment gateways and real-time inventory. **IT Officer**, INSTRUMED (2015-2019). **Graphic Designer**, freelance (2015-2017).

ALSO SHIPPED

Brawl Dinos, a live real-time multiplayer arena brawler on Google Play, built in Unity with authoritative server-side matches, real-time state synchronisation and cross-device matchmaking.

EDUCATION & AWARDS

MSc Computer Software Engineering - National University of Sciences and Technology (NUST) · **Gold Medalist**

Sep 2020 - Mar 2022

PORTFOLIO

Case studies with demo videos for every system above: haiderkamal.github.io